

RECOVERY & CNS RESTORATION

Ch 1: Recovery Hierarchy

Recovery is the process by which the body repairs, adapts, and grows stronger in response to training stress. Without adequate recovery, training is just accumulated stress — no positive adaptation occurs. Understanding the recovery hierarchy helps you prioritise correctly.

Recovery Level	Time to Recover	What Recovers	How to Accelerate
Phosphocreatine (PCr)	2–5 minutes	ATP-PCr system for explosive power	Complete rest between sets; creatine supplementation
Muscle glycogen	24–48 hours	Intramuscular energy stores	High-carb post-workout meal within 2 hours
Muscle microtrauma	48–72 hours	Structural protein repair and remodelling	Protein 2g/kg; sleep; BPC-157 peptide; cold therapy
CNS fatigue	72–120 hours	Neural drive and motor unit recruitment	Deload; sleep; stress reduction; reduce training volume
Hormonal balance	Days to weeks	Cortisol, testosterone, GH equilibrium	Consistent sleep; nutrition; stress management; adaptogens

Ch 2: Sleep Architecture for Athletes

A complete sleep cycle lasts 90 minutes. You ideally complete 4–6 cycles per night. Each stage serves distinct physiological functions that directly impact muscle recovery, fat loss, and performance.

Sleep Stage	Type	Duration	Fitness Benefit	Disrupted By
Stage 1	Light NREM	5–10 min	Transition; minor relaxation	Noise, light, stress
Stage 2	Light NREM	20 min	Heart rate drops; memory consolidation	Caffeine, alcohol, blue light
Stage 3	Deep NREM	20–40 min	Growth hormone release; tissue repair; immune function	Late alcohol, heat, stress
Stage 4	Deep NREM	20–40 min	Maximum anabolism; fat burning; cardiac repair	Chronic stress; high cortisol
REM sleep	REM	10–60 min	Mental recovery; emotional regulation; skill consolidation	Alcohol, cannabis, late-night eating

Ch 3: HRV Monitoring & Recovery Assessment

Heart Rate Variability (HRV) measures variation between heartbeats — higher variation indicates well-recovered parasympathetic dominance. Daily HRV tracking is the most objective measure of training readiness available without a laboratory.

HRV Reading vs Baseline	Recovery State	Training Recommendation
More than +10% above baseline	Excellent — very high readiness	High-intensity session; test personal records today
Within +/- 5% of baseline	Good — normal readiness	Proceed with planned session as designed
5–15% below baseline	Moderate fatigue — reduced readiness	Reduce intensity 20%; avoid maximal lifts today
More than 15% below baseline	Significant fatigue, stress, or illness	Active recovery only: walk, stretch, mobility work

Free HRV Monitoring Apps (No Device Required)

- Elite HRV — free; camera-based HRV from iPhone/Android
- HRV4Training — research-grade; camera measurement; coaches widely use this
- Garmin Connect / Fitbit — if wearing a fitness tracker; built-in HRV tracking
- Manual method: count resting HR for 60 seconds after waking; lower is generally better recovery

Ch 4: Active Recovery Protocols

Recovery Method	Mechanism	Duration	Indian Application
Light walking (LISS)	Blood flow; lactate clearance; parasympathetic activation	20–30 min at 50% MHR	Morning walk in park; evening stroll after dinner
Cold water immersion	Vasoconstriction reduces inflammation; dopamine spike	10–15 min at 10–15 degrees C	Cold bucket shower; cold bath in Indian summer
Contrast therapy	Alternating vaso-dilation and constriction; metabolic waste clearance	Hot 3 min, cold 1 min x 4 rounds	Shower temperature alternation; easy to implement
Yoga / deep stretching	Parasympathetic activation; fascial release; ROM improvement	20–40 min	Yoga Nidra; Supta Matsyendrasana; Viparita Karani
Foam rolling	Myofascial release; DOMS reduction; tissue quality	10–20 min target areas	Focus on quads, lats, thoracic spine, calves

Ch 5: Sleep Optimisation & Supplement Stack

Timing	Action	Science
3 hours before bed	Last heavy meal; switch off bright overhead lights	Melatonin production begins when lights dim
2 hours before bed	Stop screens or use blue-light filter mode	Blue light suppresses melatonin by 50%
1 hour before bed	Warm milk with ashwagandha OR chamomile tea	Tryptophan converts to serotonin then melatonin
30 min before bed	5–10 min Yoga Nidra or progressive muscle relaxation	Reduces cortisol 23%; drops core body temperature
Bedtime	Room 18–21 degrees C; complete darkness; phone outside room	Core temperature drop triggers sleep onset
Morning	Expose eyes to natural light within 10 min of waking	Anchors circadian rhythm; reduces next night sleep latency

Supplement	Dose	Mechanism	Indian Brand
Ashwagandha (KSM-66)	600mg	Reduces cortisol; improves sleep quality score by 72%	Himalaya; generic KSM-66
Magnesium Glycinate	300–400mg	GABA activation; muscle relaxation; sleep depth	NOW Foods; HealthKart
L-Theanine	200mg	Alpha-wave induction; reduces sleep onset latency	Standalone powder (inexpensive)
Melatonin	0.5–3mg (low dose best)	Sleep onset signal; circadian regulation	Any pharmacy; start with 0.5mg
Zinc (ZMA form)	30mg zinc + 400mg Mg	GH release during sleep; testosterone production	MuscleBlaze ZMA